Veganlian
A Minimalist Food Blog on AWS

Cloud-native WordPress with automatic deployment

Julian Wuntke - Dec 2025 - Neue Fische - DEHAM24

Who I am

and how I got here

- I am Julian and have a background in media & TV production.
- TV business is slowly decreasing.
- Searching for a job with the potential of learning, growing and relevant for the future.
- The CLOUD is the future.

What bothered me

and what I wanted to do differently

I love cooking but…

- Food blogs are overloaded
- Recipe hidden under clutter
- Recipes take hours to make

So I need …

- Clean, fast and minimal website
- Recipe first
- Simple, approachable Recipes

The services I used

and what they do

CloudWatch

Monitoring

SNS

Notification

VPC

Network

Load Balancer

Routing

RDS

Database

IAM

Security

Auto Scaling

Resilience

EFS

Shared Storage

Secrets Manager

Credentials

EC 2 Instances

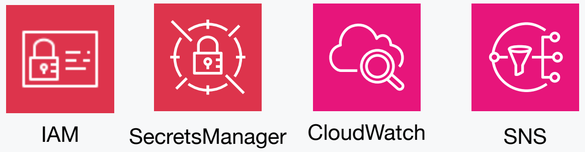
Compute

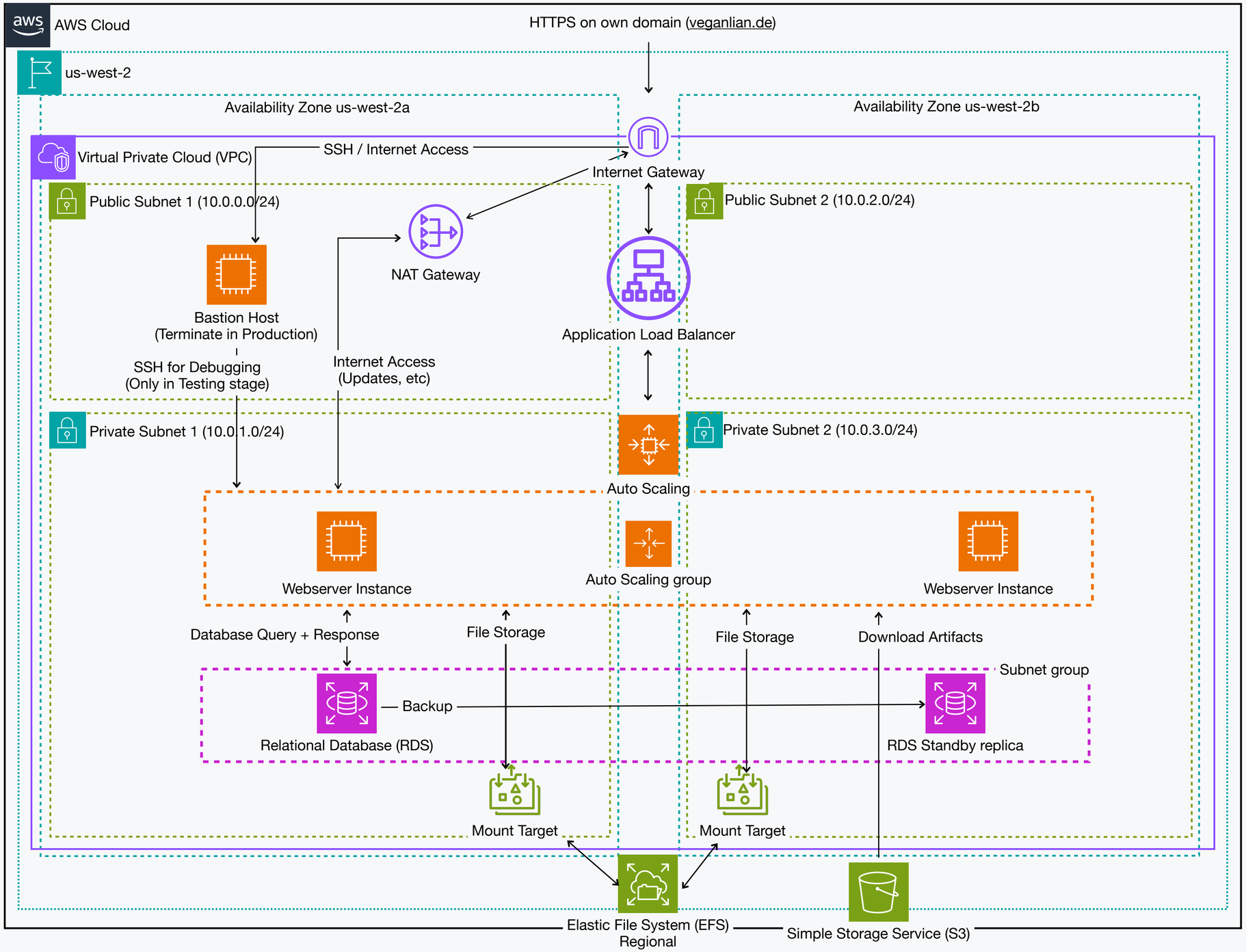
S3

Artifacts

How the system fits together

Highly available, redundant, multi-AZ solution





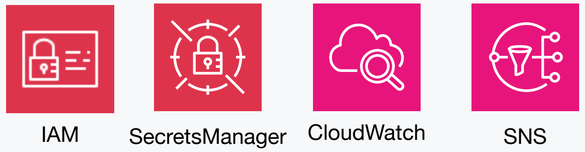
What a well-architected site needs

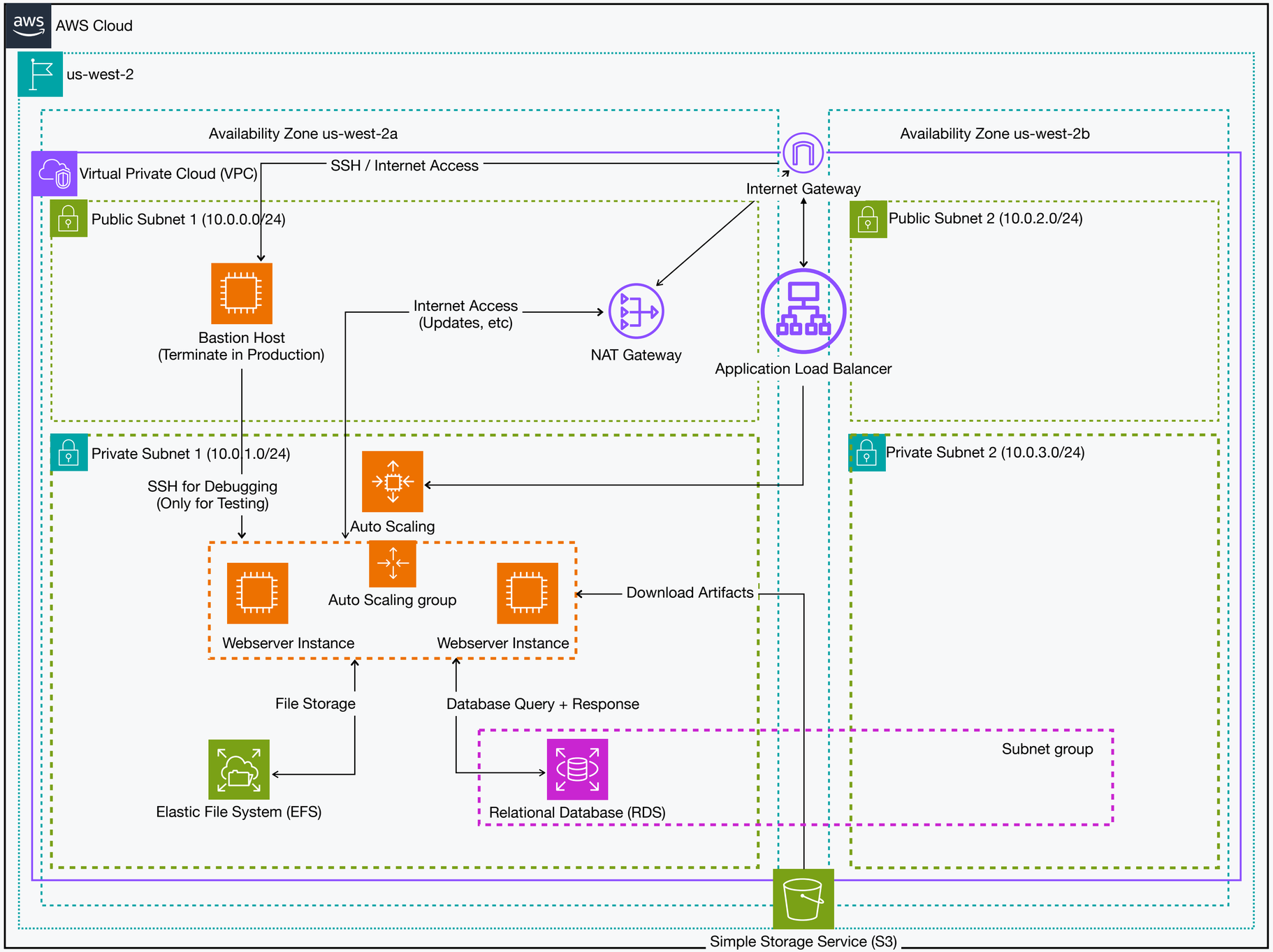
and why it matters

- **Operational Excellence** – run it as code, make changes repeatable
- **Security** – protect data and workloads on multiple layers
- **Reliability** – withstand failures and recover quickly
- **Performance Efficiency** – use the right resources and scale when needed
- **Cost Optimization** – pay for what you use instead of over-provisioning

How the system fits together

Cost-effective, highly available, single-AZ solution





What this means in costs

ALARM: "Billing Alarm"

And where the money goes

Full setup (2 AZ, all components):

Compute
- Bastion Host: t2.micro - $8.47/month (730 hours × $0.0116)
- Web Servers: 2x t2.micro (min) - $16.94/month (2 × 730 × $0.0116)
- Auto Scaling Buffer: Assuming avg 2.5 instances running
  - Additional 0.5 instance: $4.24/month (365 × $0.0116)
- Total Compute: $29.65

Database
- RDS Primary: db.t3.micro - $15.51
- RDS Standby Replica: db.t3.micro - $15.51
- RDS Storage: 20GB gp2 x2 - $4.60
- Total Database: $35.62

Storage
- EFS Regional: 50GB - $15.00
- S3 Standard: 100GB - $2.30
- EBS Volumes: 4x 8GB gp3 - $1.28
- Total Storage: $18.58

Networking
- Application Load Balancer: $22.27
- NAT Gateway: $45.00
- Data Transfer Out: 100GB - $9.00
- Data Transfer In: 100GB - $0.00 (free)
- Total Networking: $76.27

Total Large Architecture: $160.12/month

Small setup (1 AZ):

Compute
- Bastion Host: t2.micro - $8.47/month (730 hours × $0.0116)
- Web Servers: 2x t2.micro (min) - $16.94/month (2 × 730 × $0.0116)
- Auto Scaling Buffer: Assuming avg 2.5 instances running
  - Additional 0.5 instance: $4.24/month (365 × $0.0116)
- Total Compute: $29.65

Database
- RDS Single Instance: db.t3.micro - $15.51
- RDS Storage: 20GB gp2 - $2.30
- Total Database: $17.81

Storage
- EFS: 25GB - $7.50
- S3 Standard: 50GB - $1.15
- EBS Volumes: 3x 8GB gp3 - $0.96
- Total Storage: $9.61

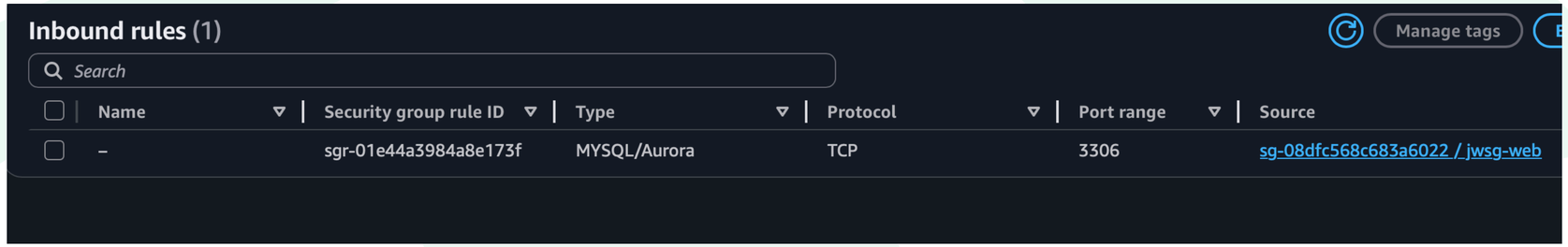
Networking
- Application Load Balancer: $22.27
- NAT Gateway: $45.00
- Data Transfer Out: 100GB - $9.00
- Data Transfer In: 100GB - $0.00 (free)
- Total Networking: $76.27

Total Small Architecture: $133.34/month

How I secure my architecture

And on which layers

- **Network layer** – isolated networks, no direct DB access from the internet
- **Identity & access** – roles and permissions instead of hard-coded credentials
- **Application & data** – only the Webserver talks to the database, secrets stored centrally
- **Encryption in transit** – all user traffic goes through HTTPS with my own domain
- **Operations** – limited admin entry points and no manual changes on live systems



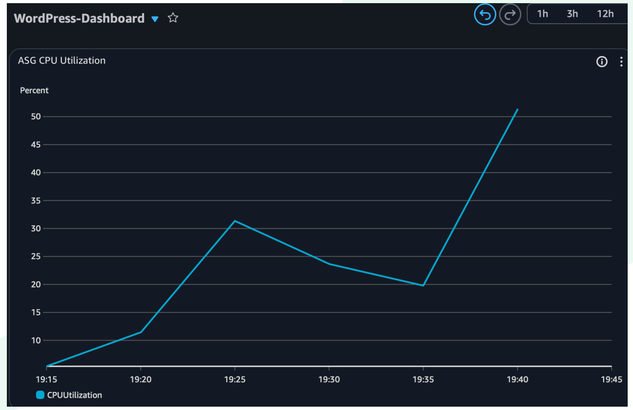
How I build and deploy it

And which tools I use

- **Mac & VS Code** – local development environment
- **GitHub** – version control
- **Terraform** – infrastructure as code and reproducible environments
- **Amazon Linux 2023** – Amazon native Server-OS
- **WordPress (with recipe plugin)** – layout, content and User Interface
- **AI tools (ChatGPT & Amazon Q)** – understanding docs, debugging, Coding

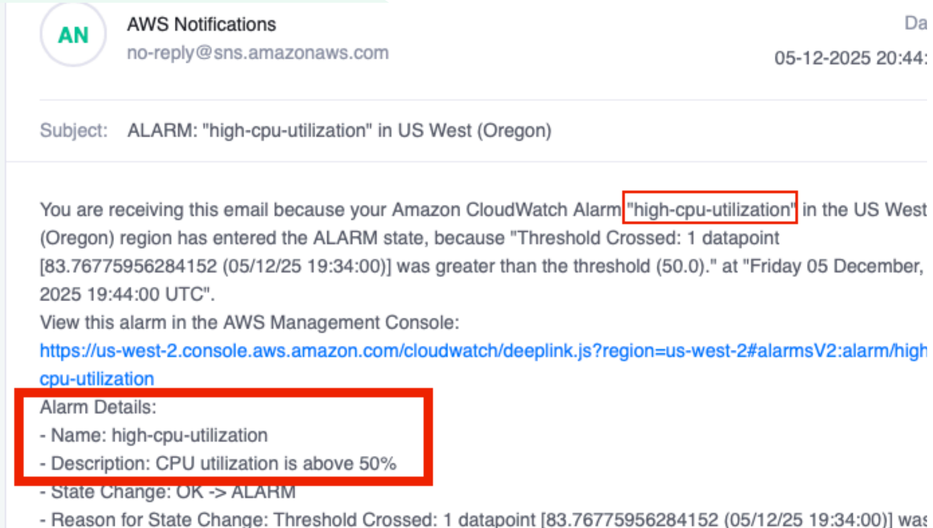
How I monitor the system

And get notified when it breaks



- **Metrics** - CloudWatch Dashboard for Monitoring Metrics
- **Logs** – saved on the Webserver
- **Alarms** – for high CPU utilization
- **Notifications** – sending alert-email

Name	State	Last state update (UTC)	Conditions
high-cpu-utilization	In alarm	2025-12-08 19:46:14	CPUUtilization > 50 minutes
low-cpu-utilization	OK	2025-12-08 19:43:12	CPUUtilization < 40 minutes



AN AWS Notifications
no-reply@sns.amazonaws.com
Da
05-12-2025 20:44:

Subject: ALARM: "high-cpu-utilization" in US West (Oregon)

You are receiving this email because your Amazon CloudWatch Alarm "high-cpu-utilization" in the US West (Oregon) region has entered the ALARM state, because "Threshold Crossed: 1 datapoint [83.76775956284152 (05/12/25 19:34:00)] was greater than the threshold (50.0)." at "Friday 05 December, 2025 19:44:00 UTC".
View this alarm in the AWS Management Console:
https://us-west-2.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-west-2#alarmsV2:alarm/high-cpu-utilization
Alarm Details:
- Name: high-cpu-utilization
- Description: CPU utilization is above 50%
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 datapoint [83.76775956284152 (05/12/25 19:34:00)] was

What makes it resilient

and why the system keeps running even if something fails

- Stateless compute
- Persistent storage
- External database
- Load Balancing
- Autoscaling
- Immutable Deploys

Live-Demo of the website

How is the user experience?

Link to Video-Presentation (without HTTPS applied, because it is only used in Capstone Presentation)

What I learned through the process

and what were the biggest challenges

- Planning an Infrastructure
- Automate everything
- How the services go hand in hand
- Cost awareness
- Real-world constraints
- In the end you'll always find a solution
- Debugging 502 Gateway
- EFS permissions
- Fetching Credentials
- Remembering what I did yesterday

What’s next

and why I should do it

- Implement more Monitoring metrics
- Centralize logging with CloudWatch Logs
- Schedule Backups for EFS and RDS
- Optimize costs (Reserved instances save 30 - 60 % costs)
- Secrets Manager rotation
- CloudFront for faster load times around the globe
- CI/CD Pipeline

Thank you for listening

and visit my socials

https://github.com/juwucloud/AWSstatelessWordpressServer

https://www.linkedin.com/in/julianwuntke/

https://veganlian.de *(available till 5 pm on 09/12/25)*